

Thermodynamics

Lecture 5A:

- Energy analysis of control volumes

Prof. Claudio Hail (he/him)

Learning objectives (I)

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Learning objectives (II)

- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation:* Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.**
- X. Knowledge: Briefly outline the history of the field of thermodynamics.

Recap Lecture 4B

Energy analysis of closed volumes

Constant volume & pressure specific heat

$$c_v = \left(\frac{\partial u}{\partial T} \right)_v \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p$$

Ideal gases

$$u = u(T)$$

$$h = h(T) = u(T) + RT$$

Integration

$$\Delta u = u_2 - u_1 = \int_1^2 c_u(T) dT$$

$$\Delta h = h_2 - h_1 = \int_1^2 c_p(T) dT$$

Averaged specific heat

$$u_2 - u_1 = c_{v,avg}(T_2 - T_1)$$

$$h_2 - h_1 = c_{p,avg}(T_2 - T_1)$$

After this lecture

- Learn how to perform a general energy analysis on a closed system for liquids and solids.
- Develop the conservation of mass principle.
- Apply the first law of thermodynamics to generalized control volumes.
- Study the relevance of flow work of fluid crossing the boundaries of a control surface.

Specific heats of solids and liquids

- Liquids and solids are often approximated as incompressible.

No difference between specific heats

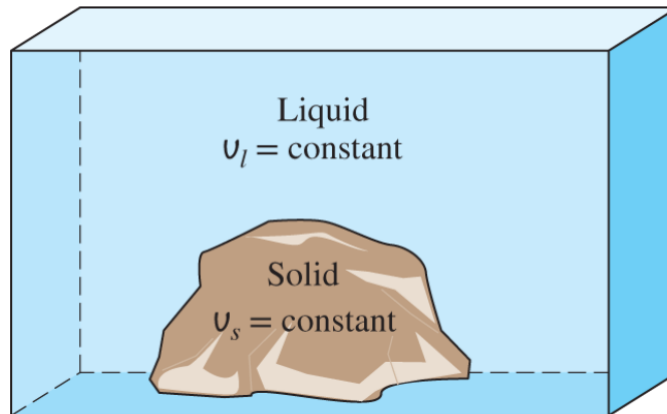
$$c_p = c_v = c$$

Tabulated values for $c(T)$

$$\Delta u = u_2 - u_1 = \int_1^2 c(T) dT$$

Small temperature differences

$$\Delta u \cong c_{avg}(T_2 - T_1)$$



Tabulated values are given in Table A-3

Enthalpy changes of incompressible substances

General enthalpy formula:

$$h = u + pv$$

Differential enthalpy:

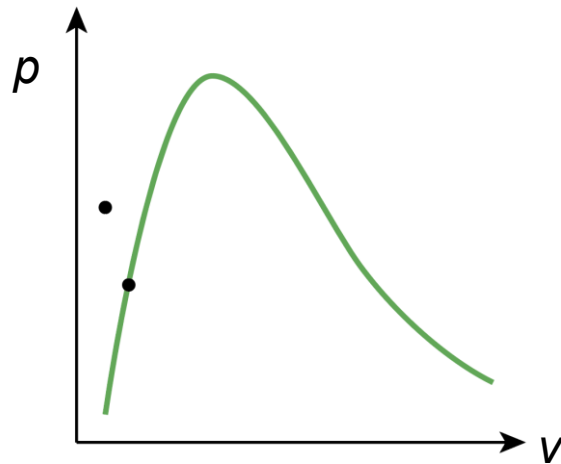
$$dh = du + vdp + pdv = du + vdp$$

Integrated: from state 1 to 2

$$\Delta h = \Delta u + v\Delta p \cong c_{avg}\Delta T + v\Delta p$$

For solids $\Delta p = 0$

Compressed liquid regime:



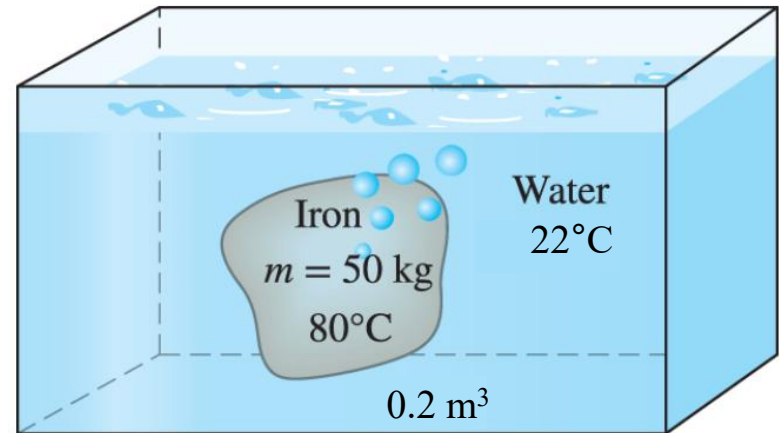
$$h(p, T) - h_{f@T} = v_{f@T}(p - p_{sat@T})$$

$$h(p, T) = h_{f@T} + v_{f@T}(p - p_{sat@T})$$

Incompressible substances: example

- An iron block at 80°C is dropped into an insulated tank of water. Determine the temperature when thermal equilibrium is reached.

Use Tables A-3a and A-3b



Incompressible substances: example

Given:

Iron block is dropped into an insulated tank of water.

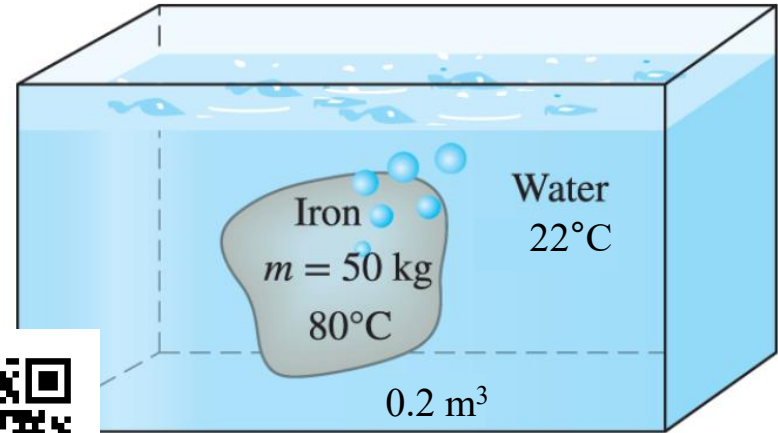
Determine:

1. The final temperature of the water and the iron block when thermal equilibrium is reached.

To Do:

- Go to pollev.com/claudiohail
- Enter the first names of your group members, e.g. Olivia and Liam.
- Submit your answers to 1, 2 and 3.

Use Tables A-3a and A-3b



Incompressible substances: example

TABLE A-3

Properties of common liquids, solids, and foods

(a) Liquids

Substance	Boiling data at 1 atm		Freezing data		Liquid properties		
	Normal boiling point, °C	Latent heat of vaporization h_{fg} , kJ/kg	Freezing point, °C	Latent heat of fusion h_{if} , kJ/kg	Temperature, °C	Density ρ , kg/m ³	Specific heat c_p , kJ/kg·K
Refrigerant-134a	−26.1	217.0	−96.6	—	−50	1443	1.23
					−26.1	1374	1.27
					0	1295	1.34
					25	1207	1.43
Water	100	2257	0.0	333.7	0	1000	4.22
					25	997	4.18
					50	988	4.18
					75	975	4.19
					100	958	4.22

Incompressible substances: example

Properties of common liquids, solids, and foods (<i>Concluded</i>)		
(b) Solids (values are for room temperature unless indicated otherwise)		
Substance	Density, ρ kg/m ³	Specific heat, c_p kJ/kg·K
Metals		
Iron	7,840	0.45
Lead	11,310	0.128
Magnesium	1,730	1.000
Nickel	8,890	0.440
Silver	10,470	0.235
Steel, mild	7,830	0.500
Tungsten	19,400	0.130

Incompressible substances: example

State 1

$$T_{1I} = 80^\circ\text{C} \quad V_W = 0.2 \text{ m}^3$$

$$T_{1W} = 22^\circ\text{C} \quad m_I = 50 \text{ kg}$$

State 2

$$T_{2W} = T_{2I} = T_2$$

Use Tables A-3a and A-3b

$$c_{pW} = 4180 \text{ J/kgK} \quad c_{pI} = 450 \text{ J/kgK}$$

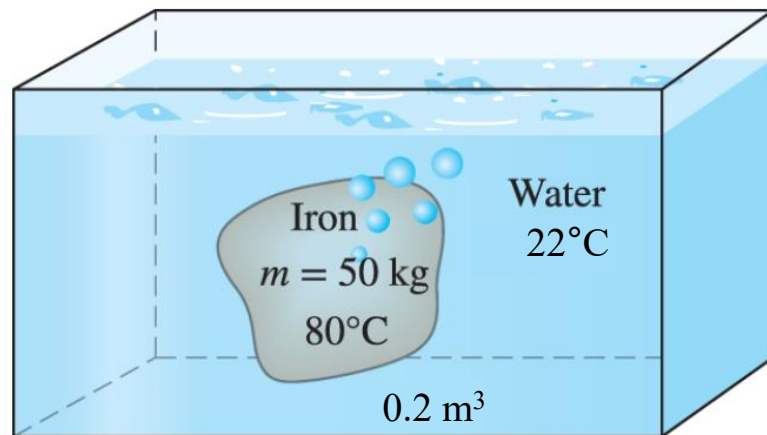
$$\rho_W = 997 \text{ J/kgK}$$

First law: $\Delta U = Q - W = 0$

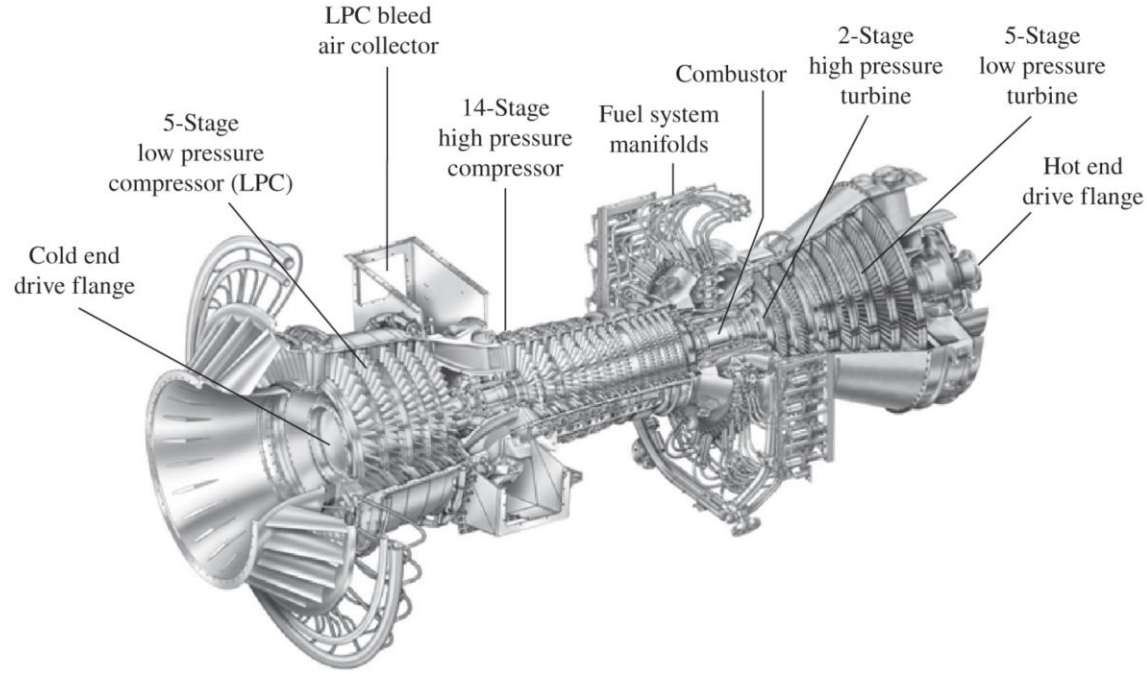
$$U_{2I} + U_{2W} - U_{1I} - U_{1W} = 0$$

$$m_W c_{pW} (T_2 - T_{1W}) + m_I c_{pI} (T_2 - T_{1I}) = 0$$

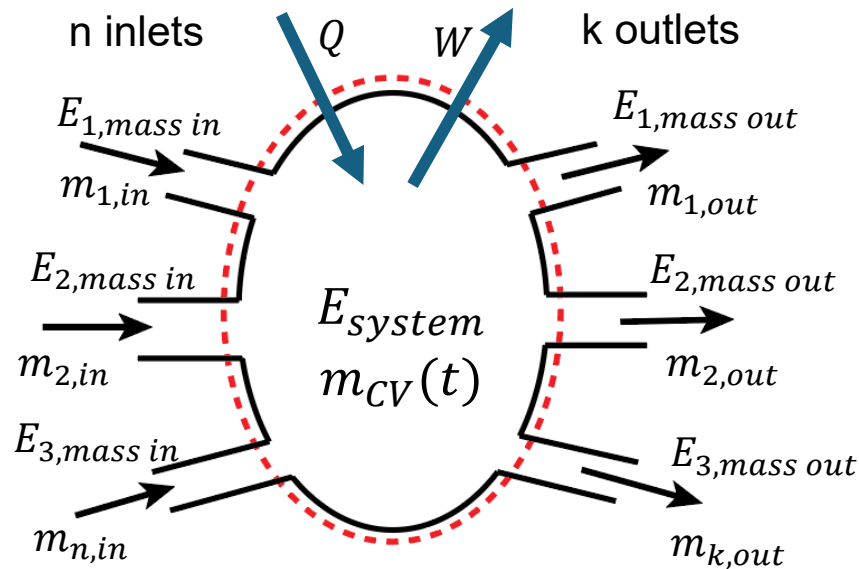
$$T_2 = \frac{m_W c_{pW} T_{1W} + m_I c_{pI} T_{1I}}{m_W c_{pW} + m_I c_{pI}} = 23.52^\circ\text{C}$$



Motivation: Energy analysis of control volumes



General control volumes



Mass flow rate

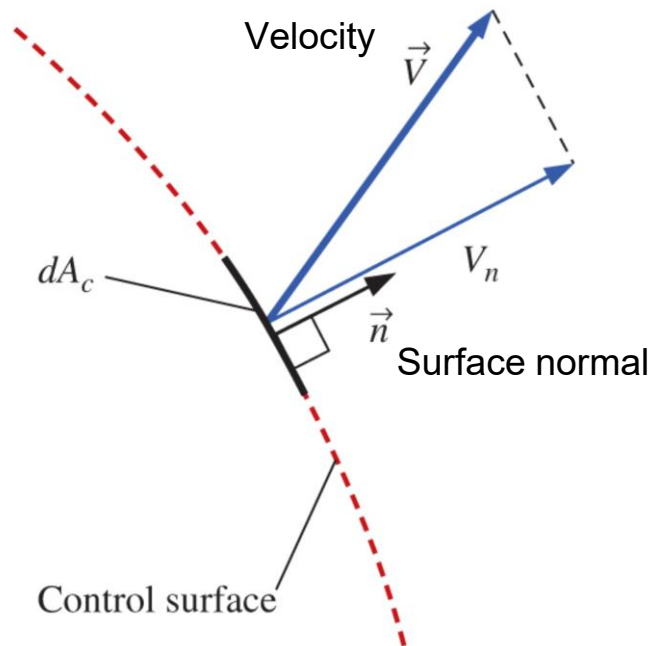
Define the mass flow rate across the boundary of the control volume

Mass flow rate (kg/s) $\dot{m}$

Differential mass flow rate $\delta\dot{m} = \rho V_n dA_c$

Inexact differential $\int_{A_c} \delta\dot{m} = \dot{m} \neq \dot{m}_2 - \dot{m}_1$

Mass flow rate integral $\dot{m} = \int_{A_c} \delta\dot{m} = \int_{A_c} \rho V_n dA_c$



Velocity profiles

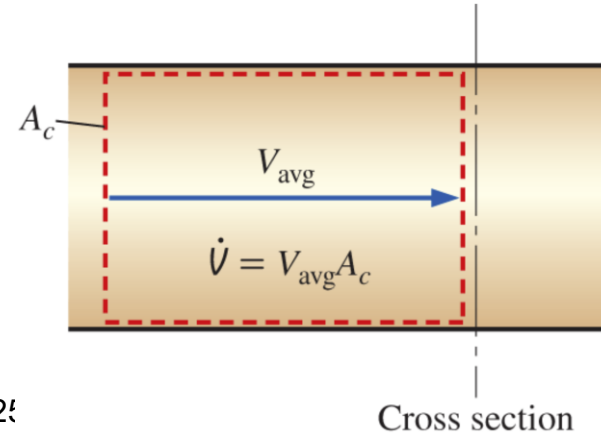
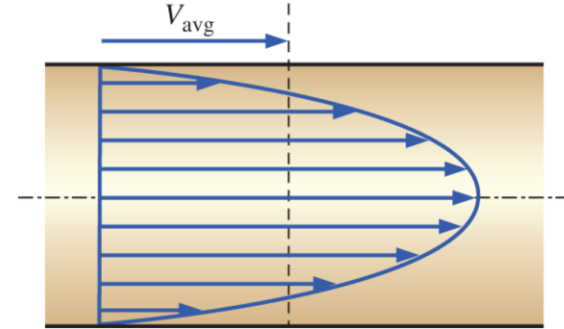
The fluid velocity in a pipe connected to a control volume is typically varying over the pipe cross section.

Fluid normal velocity $V_n = V_n(r)$

Average velocity $V_{avg} = \frac{1}{A_c} \int_{A_c} V_n dA_c$

Mass flow rate $\dot{m} = \rho V_{avg} A_c$

Volume flow rate $\dot{V} = V_{avg} A_c$



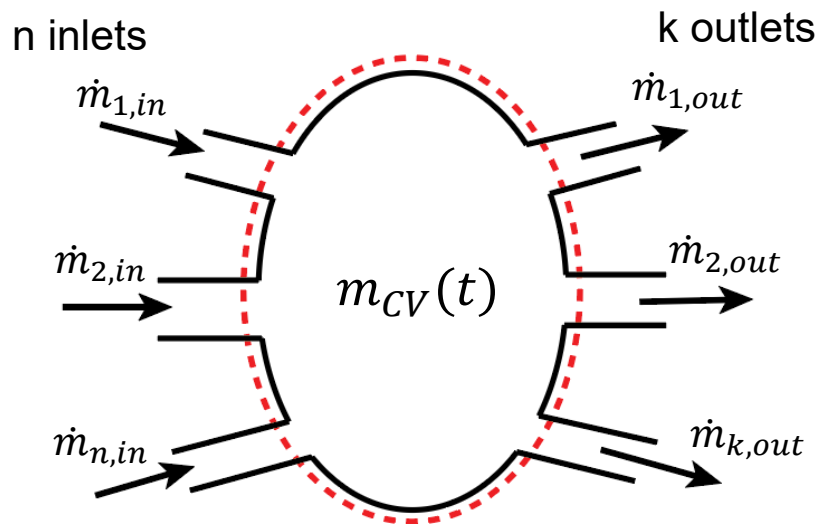
Conservation of mass

$$\left(\begin{array}{c} \text{Total mass entering} \\ \text{the CV during } \Delta t \end{array} \right) - \left(\begin{array}{c} \text{Total mass leaving} \\ \text{the CV during } \Delta t \end{array} \right) = \left(\begin{array}{c} \text{Net change of mass} \\ \text{within the CV during } \Delta t \end{array} \right)$$

$$m_{in} - m_{ex} = \Delta m_{CV}$$

$$\dot{m}_{in} - \dot{m}_{ex} = \frac{dm_{CV}}{dt}$$

$$\sum_{i=1}^n \dot{m}_{in,i} - \sum_{i=1}^k \dot{m}_{ex,i} = \frac{dm_{CV}}{dt}$$

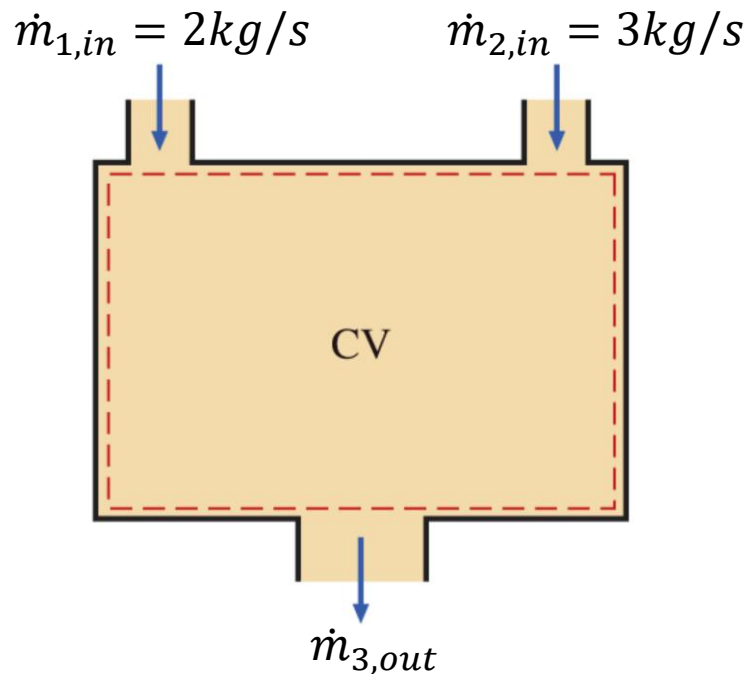


Mass balance for steady flow

$$\sum_{i=1}^n \dot{m}_{in,i} - \sum_{i=1}^k \dot{m}_{ex,i} = \frac{dm_{CV}}{dt} = 0$$

$$\dot{m}_{1,in} + \dot{m}_{2,in} = \dot{m}_{3,out}$$

$$\dot{m}_{3,out} = 5 \text{ kg/s}$$

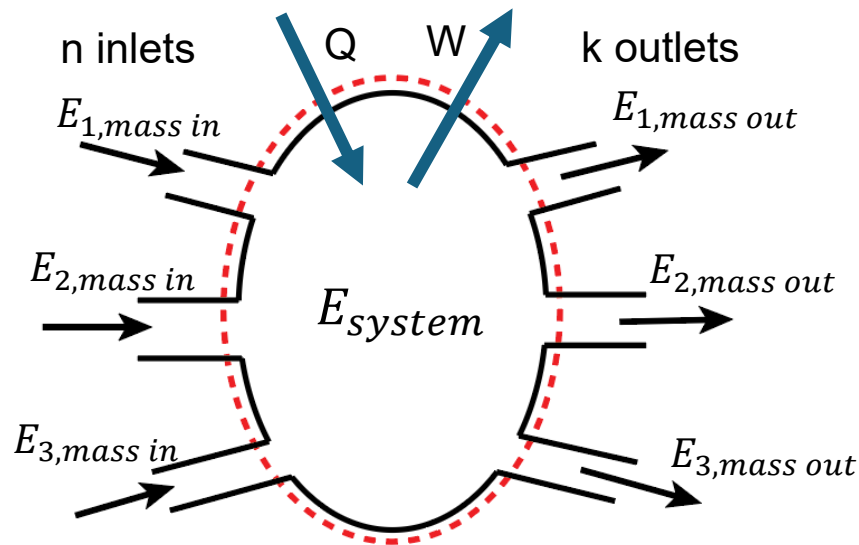


Energy balance of a control volume (I)

$$\Delta E_{system} = E_{in} - E_{out}$$

$$\Delta E_{system} = Q - W_{tot} + \sum_{i=1}^n E_{mass,in,i} - \sum_{i=1}^k E_{mass,ex,i}$$

$$\frac{E_{mass,in,i}}{m_{in,i}} = e = u + \frac{V^2}{2} + gz$$



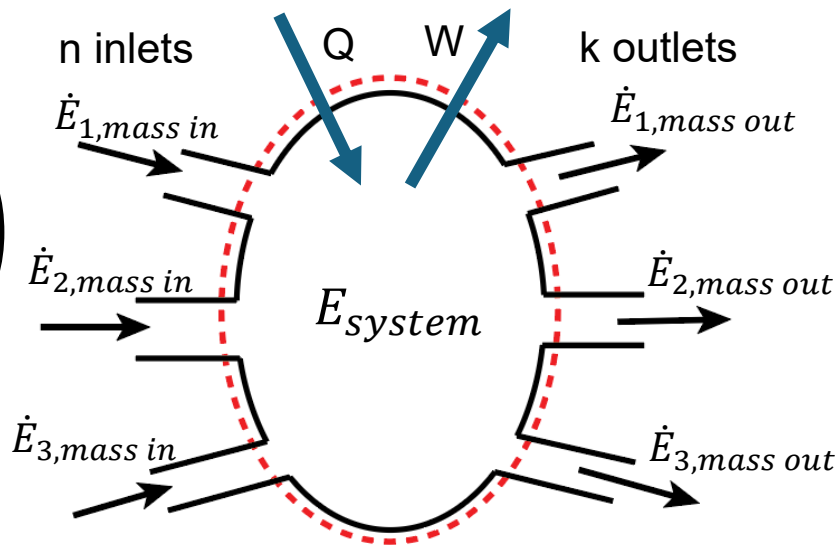
Energy balance of a control volume (II)

$$\frac{dE_{system}}{dt} = \dot{E}_{in} - \dot{E}_{out}$$

$$\frac{dE_{system}}{dt} = \dot{Q} - \dot{W}_{tot} + \left(\sum_{in}^n \dot{E}_{mass,in} - \sum_{out}^n \dot{E}_{mass,out} \right)$$

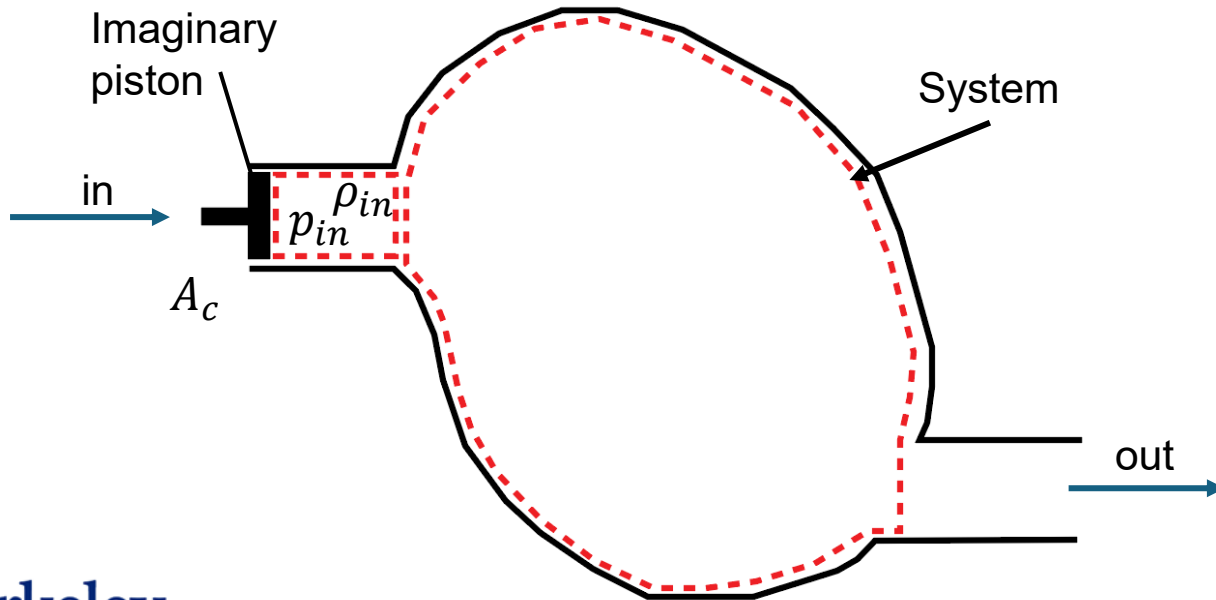
$$\frac{\dot{E}_{mass,in}}{\dot{m}_{in}} = e = u + \frac{V^2}{2} + gz$$

$$\dot{W}_{tot} = \dot{W}_s - \dot{W}_{flow,in} + \dot{W}_{flow,out}$$



Flow work/energy (I)

- Pushing fluid into a control volume requires work to be done.



Flow work/energy (II)

- Similar to work done against an imaginary piston.

$$W_{flow,in} = FL = p_{in}A_cL$$

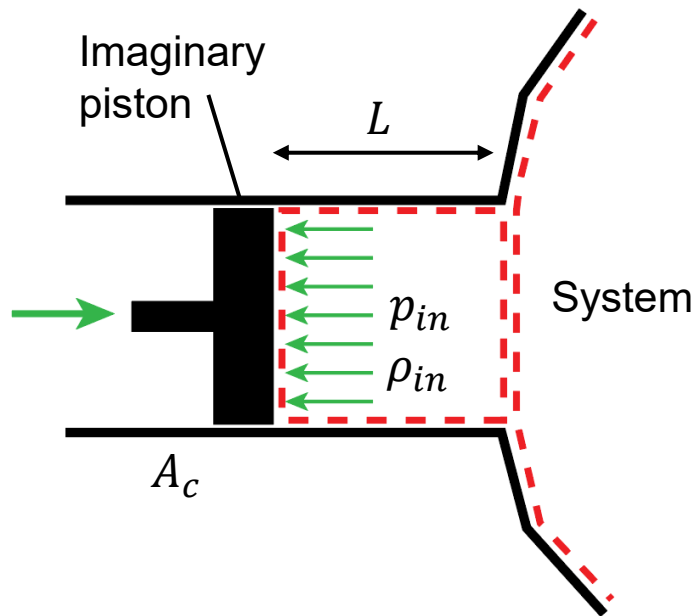
$$W_{flow,in} = p_{in}V_{in}$$

$$w_{flow,in} = p_{in}v_{in}$$

$$\dot{W}_{flow,in} = \dot{m}_{in}w_{flow,in} = \dot{m}_{in}p_{in}v_{in}$$

$$\dot{W}_{tot} = \dot{W}_s - \dot{W}_{flow,in} + \dot{W}_{flow,out}$$

$$\dot{W}_{tot} = \dot{W}_s - \sum_{i=1}^n \dot{m}_{in}p_{in}v_{in} + \sum_{i=1}^k \dot{m}_{ex}p_{ex}v_{ex}$$



Total energy of a flowing fluid

Include the flow work/energy into the total energy transport by mass.

Non-flowing fluid $e = u + ke + pe = u + \frac{V^2}{2} + gz$

Flowing fluid $\theta = e + pv = u + ke + pe + pv$
 $\theta = h + \frac{V^2}{2} + gz$

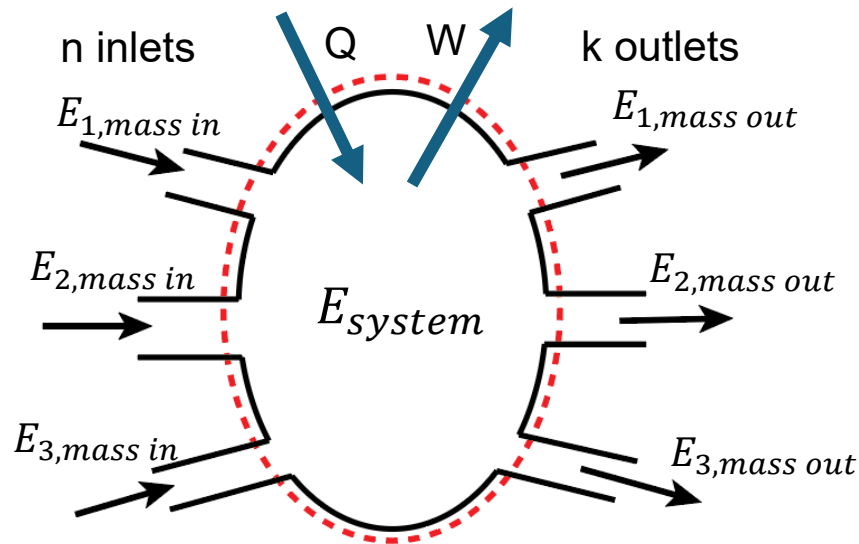
Energy transport by mass $E_{mass} = m\theta = m\left(h + \frac{V^2}{2} + gz\right)$

Rate of energy transport by mass $\dot{E}_{mass} = \dot{m}\theta = \dot{m}\left(h + \frac{V^2}{2} + gz\right)$

Energy balance of a control volume

$$\frac{dE_{system}}{dt} = \dot{E}_{in} - \dot{E}_{out}$$

$$\Delta E_{system} = \dot{Q} - \dot{W}_s + \sum_{i=1}^n \dot{m}_{in,i} \left(h_{in,i} + \frac{V_{in,i}^2}{2} + gz_{in,i} \right) - \sum_{i=1}^k \dot{m}_{ex,i} \left(h_{ex,i} + \frac{V_{ex,i}^2}{2} + gz_{ex,i} \right)$$



Summary

- Learn how to perform a general energy analysis on a closed system for liquids and solids.
- Develop the conservation of mass principle.
- Apply the first law of thermodynamics to generalized control volumes.
- Study the relevance of flow work of fluid crossing the boundaries of a control surface.

Next up

- Apply the 1st law of thermodynamics to steady-flow systems.
- Solve energy balance problems for common steady-flow devices such as turbines, nozzles, compressors, valves, mixing chambers and heat exchangers.